

Ch 5 Solution to System of Linear Equations

Solving a system of linear equations is a fundamental problem in numerical methods. These systems can be solved using **direct methods** (which provide exact solutions under ideal conditions) or **iterative methods** (which approximate solutions and are useful for large systems). Below is a detailed explanation of the methods.

Direct Methods

Direct methods aim to find an exact solution in a finite number of steps. These methods are efficient for smaller systems but may become computationally expensive for large matrices.

Gauss Elimination Method

The **Gauss Elimination Method** is used to solve a system of linear equations by converting the coefficient matrix into an upper triangular form using row operations. The solution is then obtained through **back-substitution**. **Steps:**

- Convert the system into an augmented matrix.
- Use row operations to form an **upper triangular matrix** (zeroing out elements below the diagonal).
- Solve for the unknowns using **back-substitution**.

Advantages	Disadvantages
Simple and straightforward method.	Cannot easily handle changes in system (needs full re-computation).
Good for small systems.	Sensitive to round-off errors (especially for large systems).
Always gives exact solution if no division by zero.	Computationally expensive for large matrices.

Gauss-Jordan Method

The **Gauss-Jordan Method** is an extension of Gauss Elimination where the matrix is further reduced to the **identity matrix**, eliminating both lower and upper triangular elements. **Steps:**

- Convert the system into an augmented matrix.
- Use row operations to convert the matrix into **reduced row echelon form (identity matrix)**.
- The rightmost column directly gives the values of the unknowns.

Advantages	Disadvantages
Directly gives inverse of matrix as well (along with solving $Ax = b$).	More computational steps than Gauss Elimination.
No need for back substitution step.	Higher chances of round-off errors.
Useful when solving multiple systems with same coefficient matrix.	Not efficient for large systems.

LU Decomposition Method

The **LU Decomposition Method** factorizes a matrix A into two matrices: $A=LU$, where:

- L is a **lower triangular matrix**.
- U is an **upper triangular matrix**.

Steps:

1. Decompose A into L and U .
2. Solve $LY=B$ using **forward substitution**.
3. Solve $UX=Y$ using **backward substitution**.

This method is computationally efficient for solving multiple systems with the same coefficient matrix.

Advantages	Disadvantages
Efficient when solving multiple systems with same matrix but different right-hand sides.	Needs additional storage for L and U matrices.
Reduces computational cost after decomposition.	Decomposition fails if pivot elements are zero (needs pivoting).
Good for large systems.	Initial decomposition can still be expensive.

Iterative Methods

Iterative methods are useful for solving large systems where direct methods are computationally expensive. These methods start with an initial guess and refine it through iterations.

Gauss Jacobi Method

The **Jacobi Method** solves each variable independently in an iterative manner. **Steps:**

1. Rearrange equations so that each equation solves for a single variable.
2. Start with an initial guess.
3. Compute new values of each variable using the previous iteration's values.
4. Repeat until convergence.

Advantages	Disadvantages
Easy to implement.	Converges slowly.
Parallel computation is possible (good for parallel processors).	Convergence not guaranteed (needs diagonally dominant matrix).
Suitable for simple iteration-based problems.	Not good for ill-conditioned systems.

Gauss-Seidel Method

The **Gauss-Seidel Method** is an improvement over Jacobi, where updated values are immediately used in the next calculations. **Steps:**

1. Rearrange equations for easy computation of each variable.
2. Start with an initial guess.
3. Compute new values using the latest available values.
4. Repeat until convergence.

Advantages	Disadvantages
Generally faster convergence than Jacobi.	Convergence still not guaranteed without diagonal dominance.
Easy to implement with simple code.	Cannot be parallelized easily (sequential dependency).
Good for sparse and large matrices.	May fail for badly scaled systems.

Key Difference from Jacobi:

- **Jacobi Method:** Uses previous iteration values for all variables.
- **Gauss-Seidel Method:** Uses updated values as soon as they are available, leading to faster convergence.

Direct Method vs Iterative Method

Feature	Direct Method	Iterative Method
Nature	Provides an exact solution	Approximates the solution iteratively
Computation	High computational cost for large systems	More efficient for large systems
Error Handling	Minimal rounding errors	Errors propagate across iterations
Examples	Gauss Elimination, Gauss-Jordan, LU Decomposition	Jacobi, Gauss-Seidel

Gauss Elimination vs Gauss-Jordan vs LU Decomposition

Feature	Gauss Elimination	Gauss-Jordan	LU Decomposition
Goal	Converts to upper triangular form	Converts to identity matrix	Factorizes matrix into L and U
Computation	Uses back-substitution	Uses more row operations	Efficient for solving multiple systems with the same coefficient matrix
Efficiency	Faster than Gauss-Jordan	More computational steps	More efficient for large matrices
Usage	General system solving	Inverse of a matrix	When solving multiple equations with the same coefficient matrix

Gauss-Jacobi vs Gauss-Seidel

Feature	Gauss-Jacobi	Gauss-Seidel
Update Strategy	Uses previous iteration values for all variables	Uses updated values immediately
Convergence	Slower	Faster than Jacobi
Computational Cost	More iterations required	Fewer iterations needed
Parallelization	Can be easily parallelized	Harder to parallelize due to dependency on updated values

Method	Best For
Gauss Elimination	Single small system.
Gauss-Jordan	Finding inverse + solution together.
LU Decomposition	Repeated solving with same matrix.
Gauss-Jacobi	Parallel, simple problems.
Gauss-Seidel	Faster convergence for large sparse systems.